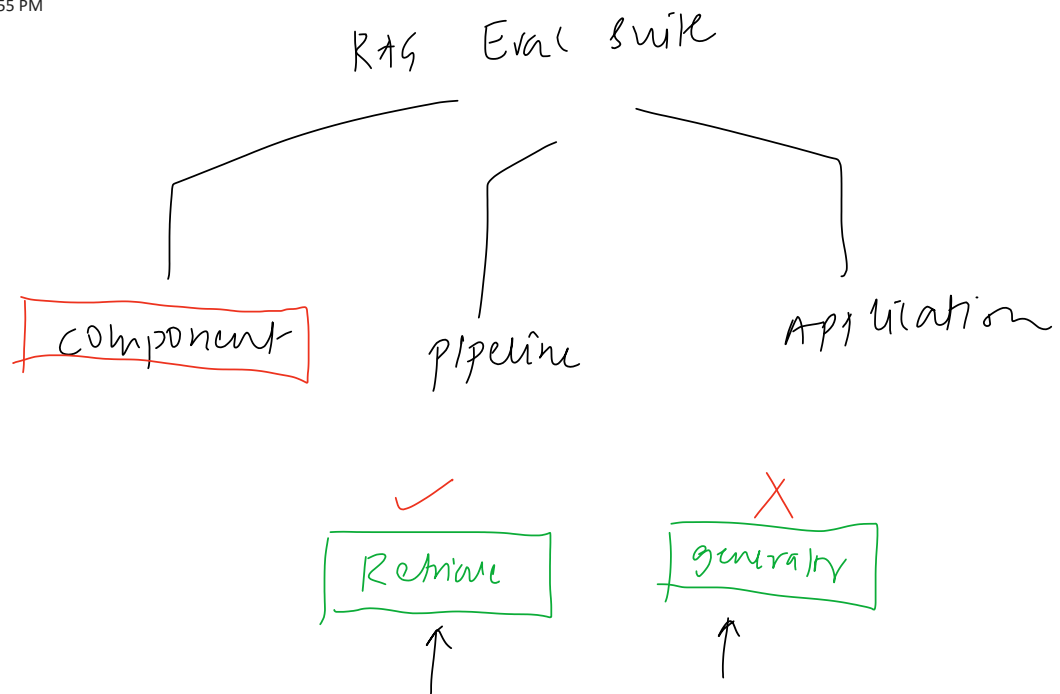


Plan of Action

Saturday, 1 August 2026 4:55 PM



Setting up the Project

Saturday, 8 August 2026 11:44 AM

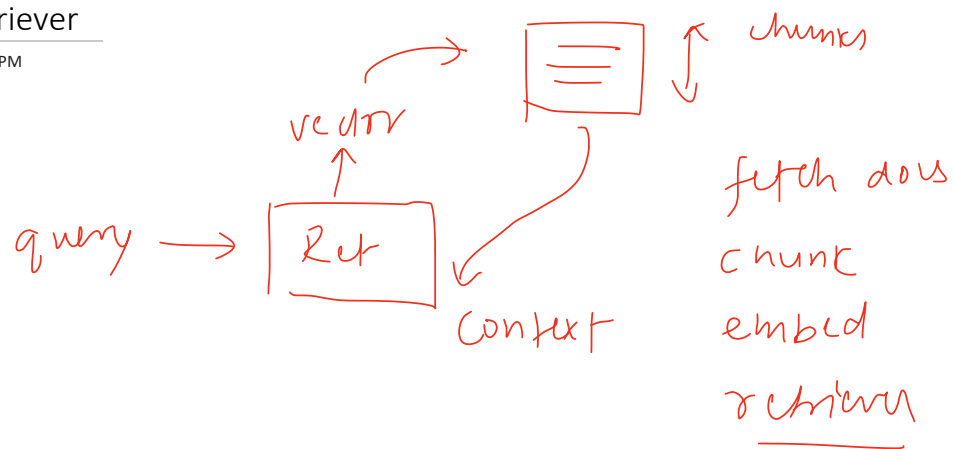
1. Create a new folder - rag eval
2. Create directory structure
3. Add transcripts
4. Create virtual env
5. Install dependencies
6. Create .env file

```
uv init --python 3.11
```

```
uv add langchain-openai langchain-chroma langchain-text-splitter:  
sentence-transformers deepeval pytest python-dotenv
```

Building the Retriever

Saturday, 1 August 2026 4:57 PM



Retriever Evaluation 1-100

Saturday, 1 August 2026 4:57 PM

(K)

tradeoff

query →

ABC → 1, 15, 13

→ 27, 28, 29

The 2 failure modes of a retriever

The tension between the 2

Why retriever eval metrics are reference based?

How do we exactly calculate recall and precision

noise
1, 15, 27, 28, 29
x x x
↓
gene

retrieval fail?

miss
(recall) ↑

(5)
(3)

→ 3/5 → 3/5

0.65

T/F

yes

(2)

noise

(precision) ↑

2/5

K=5 10

10/(5) (5) noisy

reference based evals

retrieval

recall

precision

evaluate

q → correct content

(50)

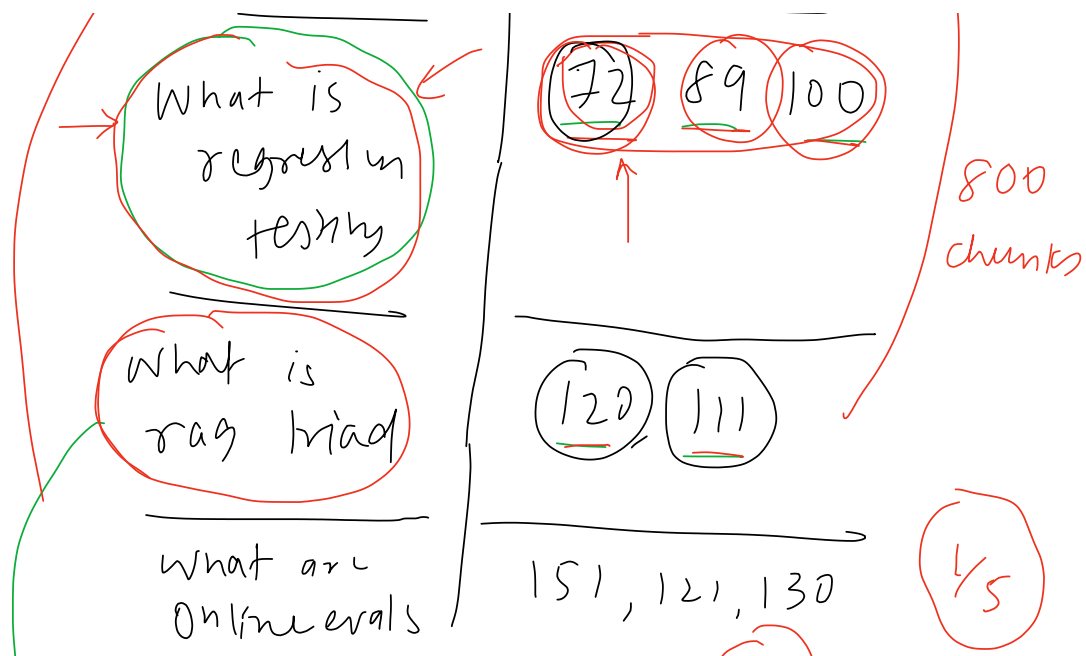
questions Golden dataset

dataset

question

doc/chunk ID

X



retriever → 1, 2, 3, 120, 5 $1/2 = 0.5$ $\frac{3}{3} = 1$ → $\frac{3}{5}$

72, 81, 89, 99, 100

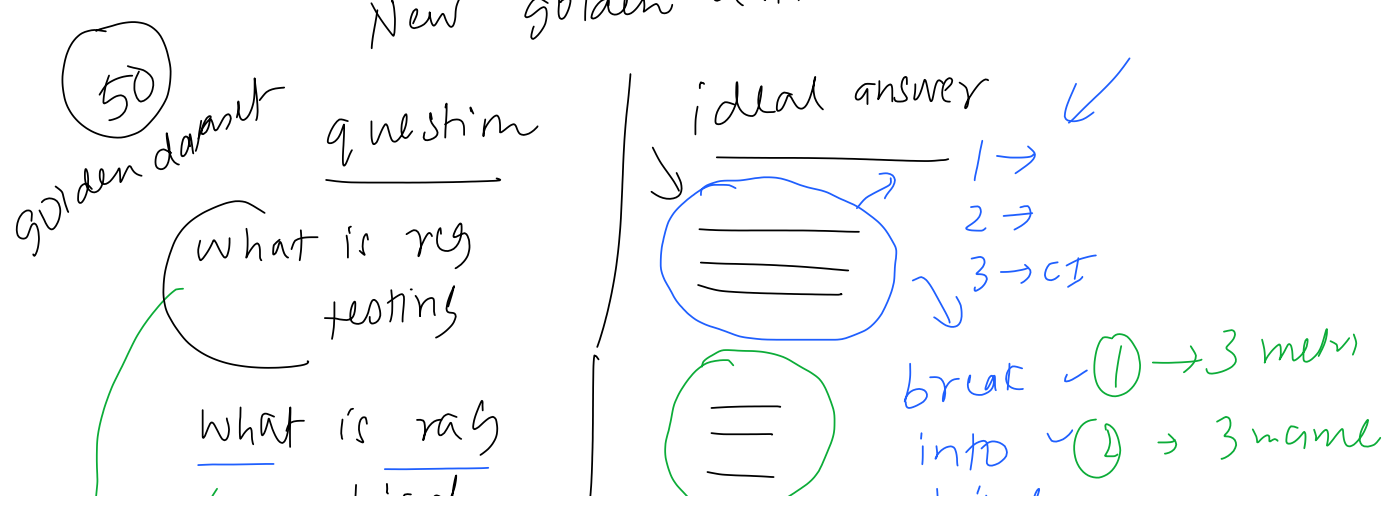
recall → 0.65 → chunking param → 750 / 100

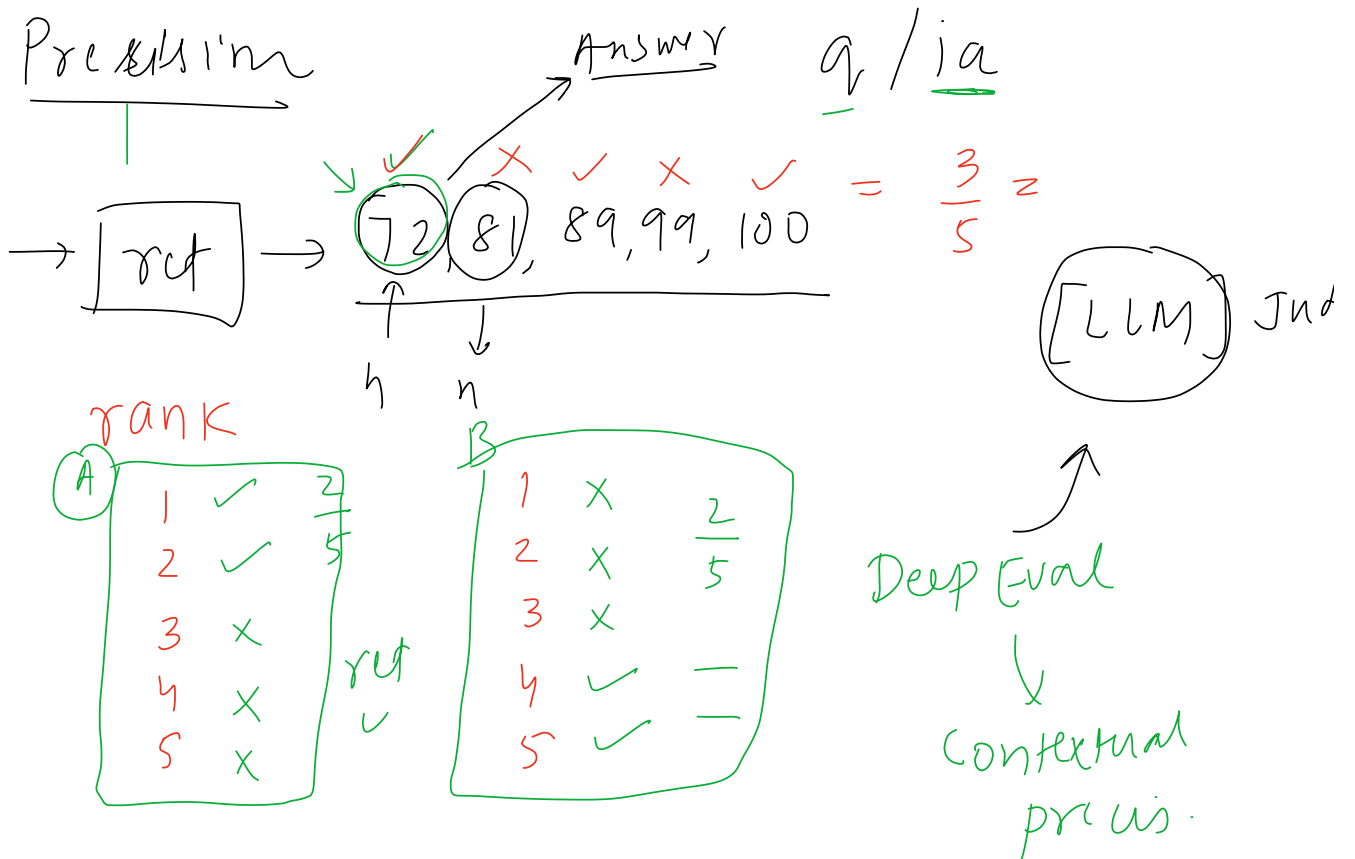
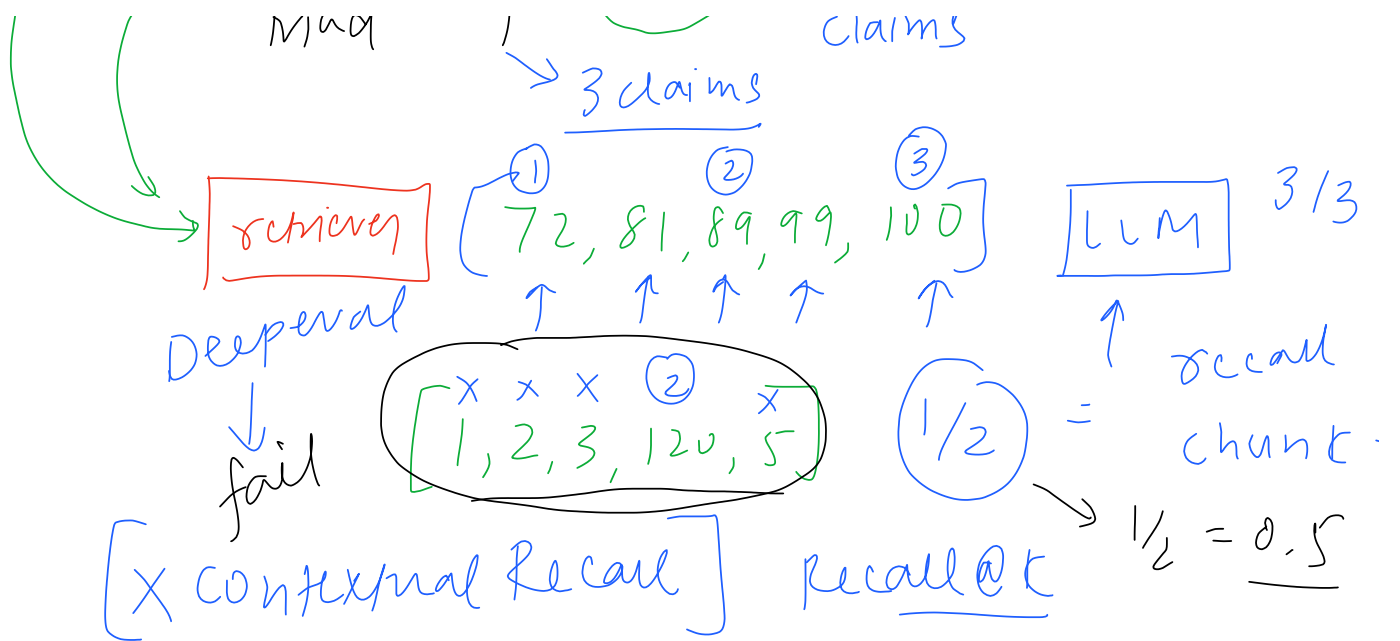
800 / 700

embeddings → 1000 / 150

doc 1 / doc 2 / doc 3

New golden dataset





A

1	✓	$\frac{1}{1} - \frac{1}{1} = \frac{1}{1} = 1$
2	✓	$\frac{2}{2} - \frac{2}{2} = \frac{2}{2} = 1$
3	x	$\frac{2}{3} - \frac{3}{3} = \frac{2}{3} = 0.66$
4	x	$\frac{2}{4} - \frac{4}{4} = \frac{2}{4} = 0.5$
5	x	$\frac{2}{5} - \frac{5}{5} = \frac{2}{5} = 0.4$

$qus = \text{---}$

B

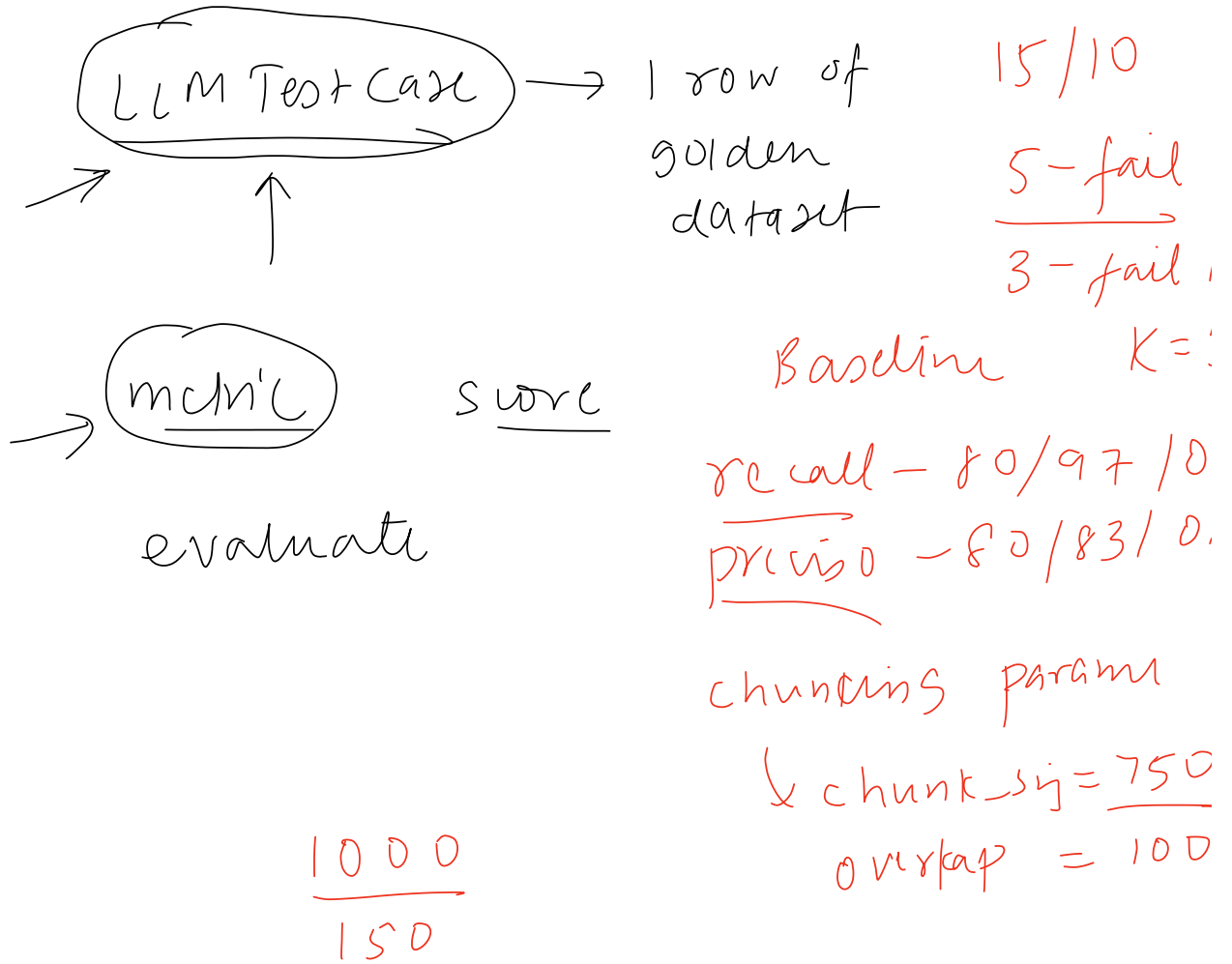
1	X	<u>0</u>	-	<u>1</u>	-	0	0/1
2	X	<u>0</u>	-	<u>2</u>	-	0	0/2
3	X	<u>0</u>	-	<u>3</u>	-	0	0/3
4	✓	<u>1</u>	-	<u>4</u>	-	1/4 = 0.25	
5	✓	<u>2</u>	-	<u>5</u>	-	2/5 = 0.40	

avg

Retriever Golden Dataset

Saturday, 1 August 2026 4:57 PM

- 1. Hand authored
- 2. LLM assisted drafting and Human review
- 3. DeepEval synthesizer and Human review
- 4. Production log mining



Improvements

Saturday, 8 August 2026 11:45 AM

1. Changing K
2. Chunking params
3. Reranker
4. Better embedding model